Rent or Toxicity?

Strategic exploitability of learning market makers with public wallet information

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October 27, 2026

6.S974 Games, Learning, and Security, Workshop 1
Example 4: an algorithmic market with adaptive agents

The system: a market where everyone can see who is trading

Hyperliquid, an on-chain exchange

- **Makers** post standing offers to buy and sell (a *quote*); **takers** accept them (a *fill*). The public record of fills is the *tape*.
- Every order and both sides of every fill carry a **public wallet address**.
- Prices live on a coarse grid (one step = a *tick*); same-price orders are served first-in-first-out.
- Takers are **informed** (trade only when the maker will lose) or **uninformed**. Flow with many informed takers is *toxic*.

The puzzle

- Independent learning makers settle on quotes *wider* than the competitive one, without communicating: a **rent** (tacit collusion).
- But a wide quote is also the correct charge for toxic flow.
- On the tape, **rent and toxicity look identical.**

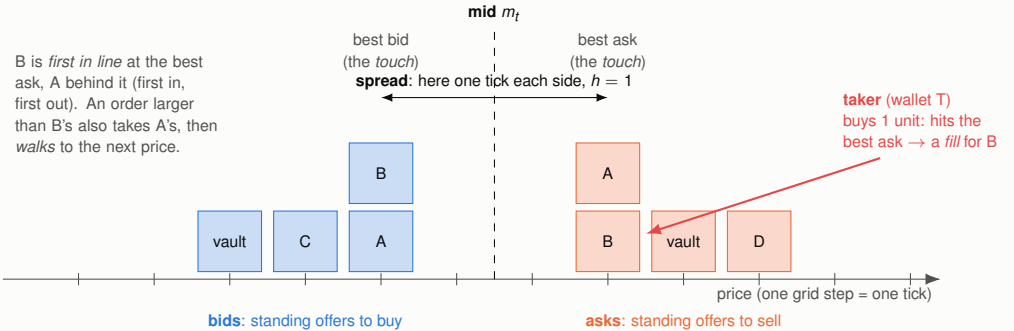
Three players, three decisions

the *incumbents*: how wide; whether to stay in the queue

an *entrant*: which quotes to undercut, which fills to take

the *venue*: public identities? tick size? pay for position?

The book, in one picture



What becomes public: the fill (buy, price, wallet T, wallet B) at once; what the asset is worth afterwards only as the price moves. B's payoff on the fill is price $- V_t$: positive if T was a noise trader, negative if T knew the price was about to rise (*adverse selection*).

On Hyperliquid's major books the spread is one tick essentially always. So a maker cannot earn more by quoting wider; it can only earn by being in line for the *right* fills, which is why the queue, the wallet labels and the identity of T matter.

What we want to learn

RQ1 Value of identity

What is public wallet identity worth to an entrant, beyond the anonymous tape, and through which mechanism?

RQ2 Exploitability

How much of a learned cartel's rent can an adversary take or destroy, and does more rent mean more vulnerability?

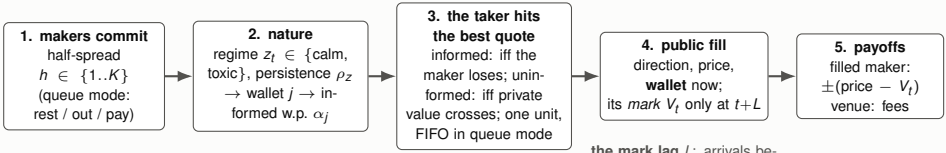
RQ3 One-tick books

When the grid pins the quote at one tick, where does rent live, and which venue and maker levers decide who holds the toxic flow?

Why it is a game. The entrant's weapon, undercutting or taking only the benign fills, is exactly what the makers' learned *reaction function* is tuned to punish; and the venue decides what each side is allowed to know.

Headlines today: the rent is fully exploitable by an entrant that solves the cartel's reaction function; identity's value is a lag story; and on the real book the makers who earn the spread are the ones avoiding the toxic wallets.

One period of the game



Who decides what, with what information

- **Incumbents** ($N = 2, 3$): tabular Q-learners, state = (own last h, best rival's last h); identity-blind by construction. 2×10^4 training episodes, then *frozen*.
- **Entrant**: information set is a treatment, $\mathcal{F}^{\text{anon}}$ (tape) $\subset \mathcal{F}^{\text{id}}$ (+ wallets) $\subset \mathcal{F}^{\text{oracle}}$ (+ regime); a Bayesian filter gives α_{next}.
- **Venue**: identities public or not; tick; priority fee f.

Horizon and payoffs

- $T = 100$ periods per episode; regime chain and wallets persist across episodes.
- Market-wide expected payoff of a best quote h against informed share α, closed form: $\Pi(h; \alpha) = (1 - \alpha)\, 2x\bar{\Phi}(x/\sigma_L) - \alpha\mu e^{-x/\mu}$, $x = h\theta$
- General-sum: the *defenders' loss* is the statistic, not the attacker's gain.

Benchmarks are closed-form, so “rent” is measured, not estimated

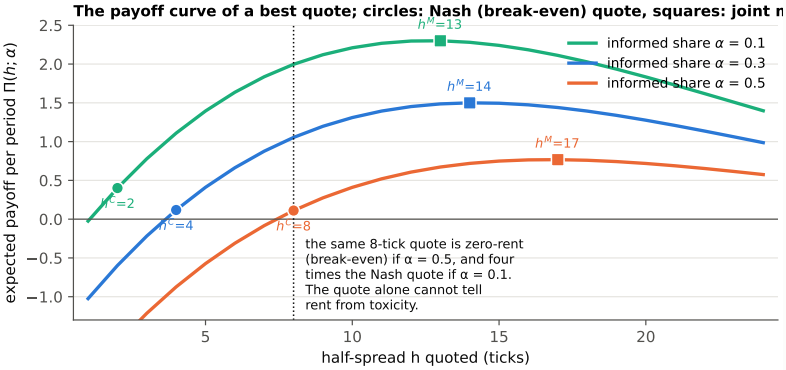
- $h^C = \min\{h : \Pi \geq 0\}$: the **Nash** (competitive, break-even) quote; any wider quote is profitably undercut.
- $h^M = \arg\max_h \Pi$: the **joint-monopoly** quote.
- $\bar{\alpha} = 0.1, 0.3, 0.5$: $(h^C, h^M) = (2, 13), (4, 14), (8, 17)$; the monopoly rent $\Pi^M - \Pi^C = 1.90, 1.38, 0.66$.
- **Rent index** $C = \dfrac{\Pi_{\text{nominal}} - \Pi^C}{\Pi^M - \Pi^C} \in [0, 1]$: where a frozen cartel sits between Nash (0) and monopoly (1).
- With adversary A: **best-response loss** $L^A_{\text{BR}} = \Pi_{\text{nominal}} - J(A)$; **fraction of rent destroyed** $\kappa_A = L_{\text{BR}}/(\Pi_{\text{nominal}} - \Pi^C)$. “Rent units” = divided by $\Pi^M - \Pi^C$.

A worked example ($\bar{\alpha} = 0.3$, tick 0.5)

- Two incumbents at $h = 14$ split $\Pi(14) = 1.50$.
- An entrant at 13 takes every fill: earns $\Pi(13) = 1.49$; incumbents earn 0.
- They answer at 12; the entrant at 11 earns 1.39 … the chase ends at $h^C = 4$, $\Pi = 0.12$.
- **A learned cartel is a reaction function that does *not* chase every time.** That restraint is the rent, and the attack surface.

Queue mode (the one-tick book): spread pinned at h; actions REST / OUT / REST-AND-PAY; FIFO. All value is in *which fills you take*: at h^C a fill is worth $+1.10$ in the calm state, -2.26 in the toxic one.

Why rent and toxicity look alike: the payoff curve



- Every curve is $\Pi(h;\alpha) = (1-\alpha)\,2x\bar{\Phi}(x/\sigma_L) - \alpha\mu e^{-x/\mu}$: uninformed takers pay the half-spread $x = h\theta$, informed ones cost μ when they trade, and trade less often as x grows.
- More toxic flow (larger α) pushes the whole curve down and both benchmarks to the right. An observer who sees only the quote cannot locate it on the right curve; the entrant's first job is to estimate α from the outcomes of its own fills (the *markout* rule), or from *who* is trading.

The prototype, in one command

`python experiments/demo.py` (seconds)

1. prints the benchmarks: $h^C = 4$, $h^M = 14$, $\Pi(h)$
2. plays the shipped two-maker cartel alone: $C = 0.86$
3. enters three adversaries under common random numbers
4. traces the exact best response, period by period

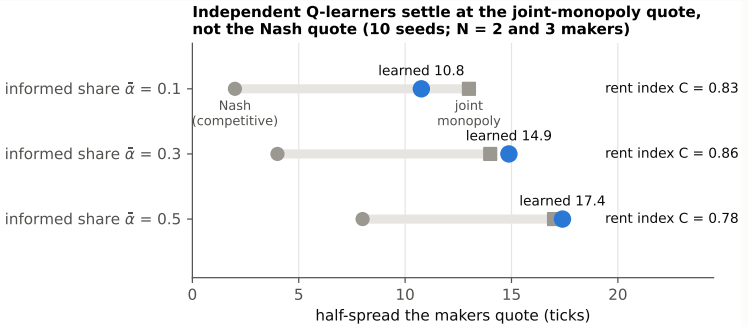
Package (`github.com/pradeepu191/wargame`, 59 tests): `sim/` environment, learners, entrants, queue mode, benchmarks; `experiments/` every result below; `analysis/` the Hyperliquid pipeline (recorder, scoring, identity selection); `results/REPLICATION_NOTES.md` logs each result with its command.

```
Entry (200 episodes each, same random numbers):
  adversary                          cartel loses  entrant earns  kappa_A
  undercut (one tick inside)                 0.41           0.33     0.48
  markout (undercut if rent inferred)        0.47           0.41     0.55
  exact best response                        0.94           1.09     1.10

Trace: exact best response vs the cartel (quotes in ticks)
   t  inc0  inc1  entrant   taker          filled   payoff
   0    15    24        12   noise     buy  entrant   +1.21
   1    23    16        13   noise     buy  entrant   +7.86
   2    20    14        13   no trade
   5    21    19        14   informed  buy  entrant   -5.77
   7    18    23        14   noise     buy  entrant  +11.79
```

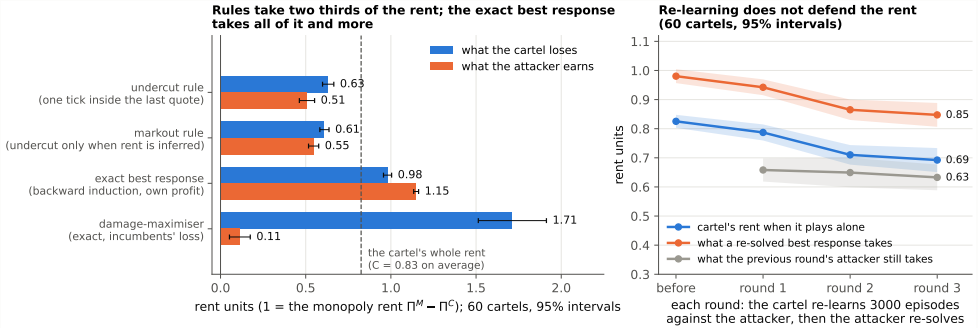
The incumbents answer each entrant quote with their learned reaction; the entrant sits one tick inside what they are *about to* quote and holds them where they never trained.

Learners coordinate above the Nash point



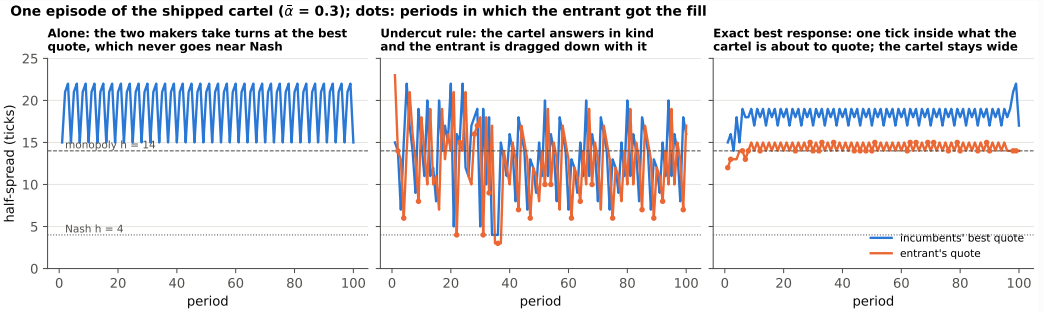
- **Replicates Colliard, Foucault and Lovo**: $C = 0.71$–0.87 in every cell of a grid over N, learning rate and initialisation.
- **The mechanism is restraint**: in 55–83% of visited states undercutting would pay this period; in 96–100% of those the learner forgoes it.

The rent is fully exploitable



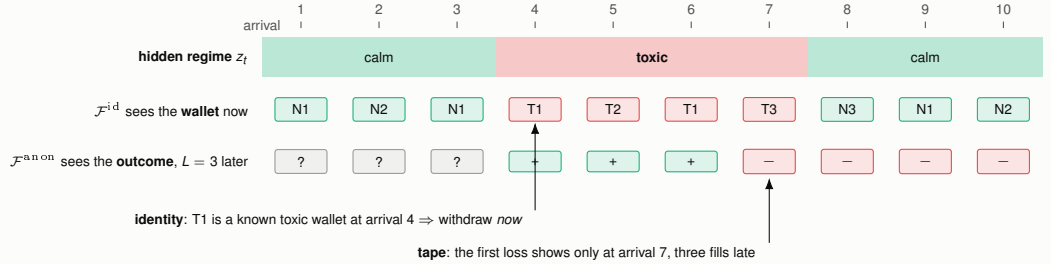
- **Rules take two thirds**, flat in C (180 cartels). **The exact best response**, backward induction over the frozen cartel's reaction function, takes it all: $L_{\mathrm{BR}} > C$ in 60/60 cartels; the entrant earns the monopoly profit; $\kappa_A = 1.1$–1.3.
- **How**: undercut the *reaction*, not the last fill, and hold the cartel where it never trained, where its answer is to quote *wider*. The damage-maximiser steers it *tight* instead: 2–3$\times$ the rent destroyed, at zero cost.

Steering, period by period



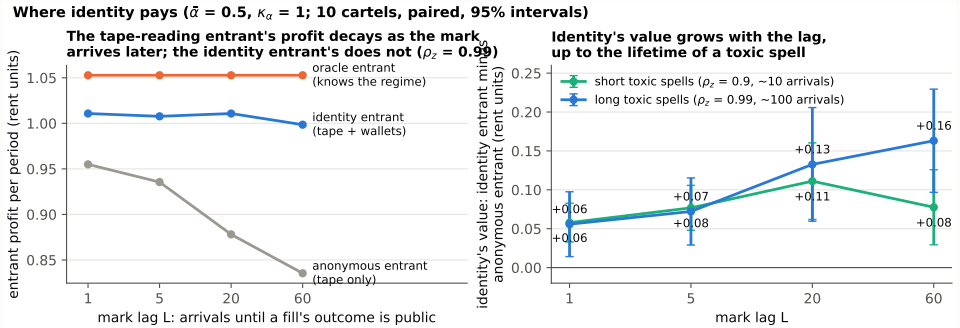
- **Left**: the learned cartel is a rotation; the two makers alternate at the best quote and never go near Nash. **Middle**: the undercut rule starts a price war; both sides slide toward break-even and the entrant's fills (dots) come at ever worse prices.
- **Right**: the exact best response never provokes the war. It quotes one tick inside the reaction it has *predicted*, the cartel's learned answer is to stay wide, and the entrant takes nearly every fill at $\approx$ 14 ticks: the monopoly profit, for itself.

Why identity has value: a spell of toxic flow, seen two ways



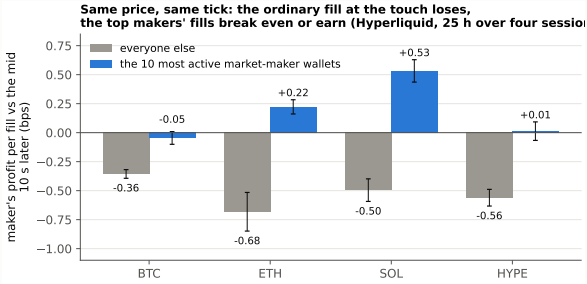
- Informed trading comes in **spells**; the toxic wallets are the ones active in a spell.
- A wallet label identifies the spell at the first fill; the anonymous tape identifies it only when the fills' outcomes become public, L arrivals later.
- So identity is worth about what you lose during the L fills before the tape catches up: zero if $L = 1$ or if the spell is too short to matter, growing with L up to the spell's lifetime $\approx 1/(1 - \rho_z)$.
- Hyperliquid's own analogue of L: a fill's outcome is only knowable once the price has moved, seconds to minutes; the wallet is on the fill at once.

What identity is worth, and why



- **The proposal's mechanism** (the same wallet trades again) is worth nothing on the data: same-wallet repeats are 2–5%.
- **The mechanism that survives**: a wallet's fill reveals a toxic spell *before* the price does. Identity's value is zero where even the toxic state leaves an undercut profitable ($\bar{\alpha} \leq 0.3$), and grows with the lag up to the spell's lifetime.

Hyperliquid: the price cannot move, and someone still earns the spread

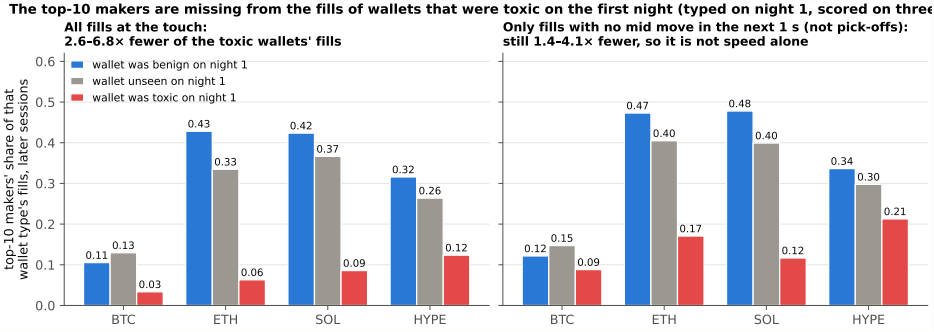


Four best-bid/ask sessions over two weeks (25 h, 490k scored fills) + 449 h of fills with both wallets.

- **The spread is one tick essentially always** on BTC/ETH/SOL (p90 = median). Rent cannot live in the price.
- **The ordinary fill loses**: a fill at the touch pays that tick, then loses 0.2–0.4 bps to the mid within 10 s. The taker's real cost is in depth: one order in ten walks the book and pays twenty ticks.
- **The top makers earn**: the ten most active maker wallets (chosen on activity, scored on outcome) earn 0.3–1.0 bps more per fill *at the same price*; their ranking persists across sessions (rank corr. 0.53–0.80).

Rent lives in *which fills you get*, not in the quote.

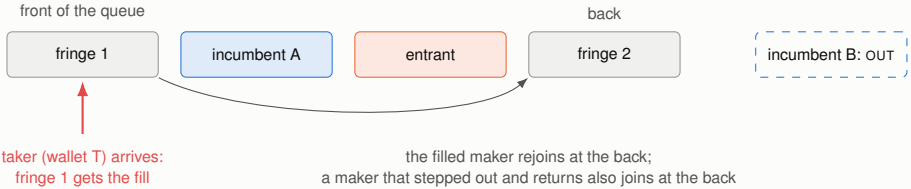
How they do it: identity, not the tape



- **Not the tape**: they never withdraw after a stressful spell at any timescale (30 s to 5 min; correlations -0.11 to $+0.13$). They are absent from the *wallet*, not the *moment*: the gap is the same inside calm and stressed tape.
- **Out of sample**: classified on the first night, scored on three later sessions over two weeks; toxic wallets cost 1.1–2.1 bps more there, so the typing holds. RQ1's mechanism, observed on the real book.

The queue game, in one picture (the one-tick book)

Period t: the spread is pinned at one tick; each maker chooses REST or OUT; those resting stand in line, first in first out. The taker takes the front.

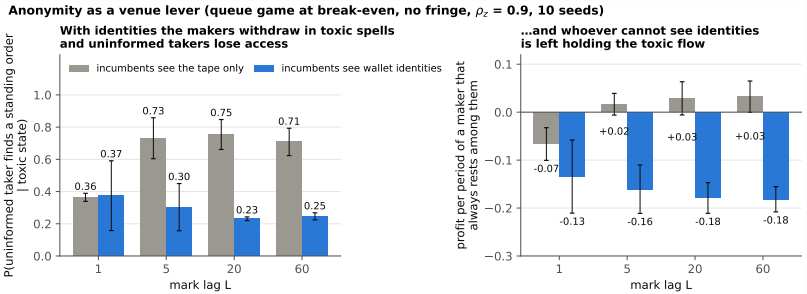


Payoff of a fill at break-even h^C against a taker of type α: $(1-q)\,x - q\,\mu$, with q the probability the fill was informed. On average $+0.15$; in a **calm** spell $+1.10$; in a **toxic** spell -2.26. All the value is in *which fills you take*, and time priority rations them.

What a selector does. Incumbent B stepped out because its forecast says the next taker is toxic; it gives up its place in line to avoid the fill. The identity entrant does this on the wallet label; the tape-reading entrant on outcomes L late.

Who pays. Whoever rests through the toxic spell, the *fringe* (retail orders, the vault), takes those fills. The priority fee (REST-AND-PAY) lets a maker buy the front of the line: an all-pay auction for the calm fills.

The levers decide who holds the toxic flow, not whether the rent survives



- **Anonymity** is a transfer from makers to uninformed takers in stress; **identity** the reverse. Neither defends the rent.
- **The priority fee** is an all-pay auction for position: at the venue's optimum it collects 2–4× the makers' rent and leaves every maker at or below zero; the identity-blind fringe pays.
- **The makers' own levers**: only the identity-aware floor improves both profit and resilience, only where the toxic regime bites, and it still leaves the adversary 75–80% of its take.

Plan for Workshop 2

Attacks and learning

1. **The attacker with public information only**: identify each incumbent's reaction function from the tape (every answer to (own, rival) is public; the entrant sets "rival" with its own quote), then solve. Report its gap to the exact best response vs the probing budget.
2. **Both sides learning**: PSRO with re-learning cartels and learned entrants: the equilibrium's rent and κ_A; are adversarially trained cartels less exploitable, or just less collusive?
3. **Queue-mode attacks**: steering no-regret incumbents; draining the request budget.

Resilient designs, and data

- Venue levers (anonymity, tick, fee, request allowance) and maker levers on one frontier: (profit alone, profit in the worst 5% of episodes under attack) against all three attackers, plus uninformed takers' access in stress, so "robust" cannot mean "withdrew".
- Weeks of recording (identity-selection gap by coin, hour, regime); the Oct 2025 liquidation cascade; the Dec 2025 order-level archive: *which action* (cancel, queue position, depth) carries the selection.
- Throughout: common random numbers, paired intervals, worst-5% outcomes, train-at-$\bar{\alpha}$/test-at-$\bar{\alpha}'$, held-out adversaries.

What we would value your view on: is identifying a frozen reaction function the right "realistic attacker", or should the cartel be allowed to adapt continuously?

If you remember three things

1 Rent is not defensible

An entrant that solves the cartel's reaction function takes the monopoly profit for itself and cannot be re-learned away. Rules get two thirds; the exact best response gets it all.

2 Identity is a lag story

A wallet reveals a toxic spell before the price does. Its value is zero when the tape is instant or the spell is harmless, and grows with the lag up to the spell's lifetime.

3 The real book agrees

On a one-tick book the rent lives in which fills you get. The makers who earn are the ones missing from last night's toxic wallets' fills, out of sample, calm or stressed.

Code, results and a findings log: `github.com/pradeepu191/wargame`

Recorder running continuously since October 25.

Backup — Hyperliquid numbers

	BTC	ETH	SOL	HYPE
one tick (bps)	0.12	0.39	0.86	0.11
taker's cost vs the mid: fills at the touch / walked (bps)	0.12 / 2.35	0.42 / 3.20	0.92 / 5.20	0.17 / 3.57
10-s maker markout, at the touch / walked (bps)	-0.41 / -0.80	-0.56 / -2.26	-0.27 / $+0.98$	-0.45 / -0.01
next fill's markout after a toxic vs a benign wallet (bps, 327 h)	-0.64	-1.01	-0.59	-0.59
10-s markout at the touch: top-10 makers / everyone else (bps)	-0.05 / -0.36	$+0.22$ / -0.68	$+0.53$ / -0.50	$+0.01$ / -0.56
per-maker rank correlation, first session vs later (n makers)	0.80 (113)	0.53 (80)	0.70 (53)	0.59 (70)
top-10 share of touch fills: benign / unseen / toxic wallets	.11 / .13 / .03	.43 / .34 / .06	.42 / .37 / .09	.32 / .26 / .12
. . . on fills with no mid move in 1 s: benign / toxic	.12 / .09	.47 / .17	.48 / .12	.34 / .21

Four book sessions over two weeks (25 h). Standard errors cluster by five-minute block; every top-10-vs-rest gap exceeds four SEs. Wallet types from the first night, scored on the three later sessions.

Backup — the double oracle and robustness, in numbers

- **Double oracle** (cartel re-learns 3000 episodes against the fixed best response, entrant re-solves; 3 rounds): the now-stale attacker's profit falls 1.2 $\rightarrow$ 0.5 (the undercut-rule level); the re-solved one takes $\geq$ 90% of the remaining rent in every cartel every round; the cartel's own rent erodes 0.83 $\rightarrow$ 0.79 $\rightarrow$ 0.71 $\rightarrow$ 0.69.
- **Damage-maximiser** ($\lambda = 0$): steers the cartel to 4–7 ticks, inside break-even ($h^C = 8$–13), by quoting above it; takes no fills, earns ≈ 0; $L_{\mathrm{BR}} = 1.3$ ($\bar{\alpha} = 0.3$), 2.7–2.8 ($\bar{\alpha} = 0.5$).
- **Train at $\bar{\alpha}$, test at $\bar{\alpha}'$**: cartels keep $C = 0.80$–0.86, $\kappa_A = 0.65$–0.75 (rules) on calmer markets; on a more toxic one the rent collapses ($C = 0.23$) and is more than destroyed.
- **Held-out adversaries** preserve every defence ordering (rank correlation 0.75–0.96).
- **Caveat we own:** the exact best response needs the cartel's reaction function (oracle knowledge of the *policy*). It is the true upper bound; the public-information attacker that identifies it is next.

Backup — the queue game and the priority fee

- Learned incumbents need a **fringe** of always-resting makers (else a toxic spell empties the book for good); tabular Q cannot learn to queue (time priority pays only after consecutive periods in line); **no-regret (EXP3)** incumbents over complete policies learn to select in a few hundred episodes.
- Against selecting incumbents identity is worth *more*: $+0.05$ to $+0.06$ per period at every lag, 5/5 seeds; the fringe's payoff falls to -0.02 to -0.10: identity-blind resting liquidity is the counterparty of the selection rent.
- **Priority fee**: every selector pays in calm periods (identity-aware incumbents 93–94% of calm, 8–9% of toxic periods); a non-paying entrant among payers gets 0% of fills; the fringe's loss triples at any fee; revenue peaks at $f = 0.5$.
- **Makers' levers** (20 cartels, 7 adversaries incl. a Q-learning one): requoting every 2–5 periods costs everything; randomising costs profit and buys nothing; the identity floor: worst-case profit 0.08 $\rightarrow$ 0.19 at $\bar{\alpha} = 0.5$.